

Gricea: A platform for Reproducible Conversational AI Research at Scale

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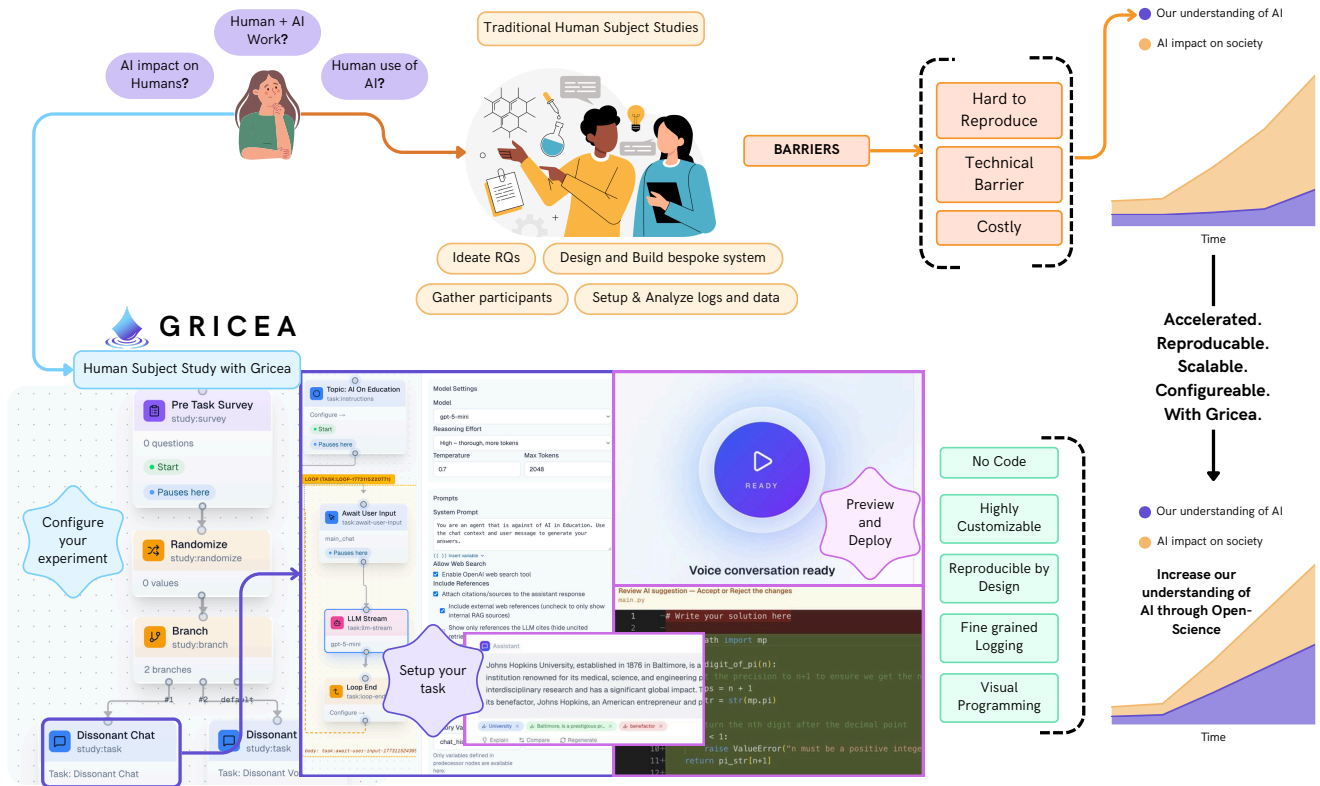


Figure 1: Overview of current challenges and opportunities with a research platform. Top: Many HCI research questions require testing *pipeline variants*—changing not just the prompt, but the interface and the agent pipeline. Because commercial systems are black-box and fast-changing, researchers often re-implement bespoke prototypes for each research question, leading to high cost, hindered reproducibility, and limited standardization. Bottom: Our research platform makes these variants configurable as study conditions: researchers can author fine-grained study procedures and task interfaces in a unified authoring environment, validate configurations, publish immutable versions, and run studies at scale through an instrumented participant runtime.

Abstract

Conversational AIs (CAIs) are reshaping how people search, communicate, and work. However, our understanding of how CAIs shape user behavior remains limited. To inform CAI design, we often need well-designed human subject studies. In practice, however, such studies are costly, technically demanding, and difficult

to reproduce. We present Gricea, a platform for designing, deploying, and analyzing reproducible CAI studies at scale. Gricea is informed by a formative analysis of prior CAI research and the broader manipulation space these systems afford, motivating support for precise experimental control, flexible multi-stage protocols, and fine-grained behavioral logging. Beyond lowering technical barriers, Gricea supports open and reproducible science by making study designs easier to inspect, share, replicate, and build upon. We evaluate Gricea with researchers and AI practitioners of diverse backgrounds, where participants create studies using the platform and reflect on its usability, and support for reproducible research through surveys and semi-structured interviews. The evaluation demonstrated that Gricea enabled participants from varied disciplinary backgrounds to construct study designs addressing a range of research questions.

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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **HCI design and evaluation methods**; **User studies**; **Empirical studies in HCI**; **Collaborative and social computing**; **Field studies**; **Usability testing**; **User models**; **Natural language interfaces**; **Web-based interaction**; **Collaborative interaction**; **User interface management systems**; **Interaction design process and methods**; • **Information systems**;

Keywords

Conversational AI, Controlled Studies, Human Subject Studies, Research Platform

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1 Introduction

Conversational AI systems built on large language models are rapidly becoming a default interface for information access, sense-making, communication, and everyday work [9]. As these systems move from novelty to infrastructure, they increasingly shape how people ask questions, evaluate evidence, revise beliefs, and act on the information they receive. This makes conversational AI an important object of human-centered study. To design, govern, and deploy these systems reliably, we need empirical methods that can rigorously examine how they shape human behavior and perform in real-world interactions.

Prior human–AI interaction research shows that outcomes depend not only on model capability, but also on how systems communicate uncertainty, present evidence, structure initiative, support verification, and distribute control between human and AI [3, 23, 33, 40]. Harms such as misinformation exposure, overreliance, persuasion, and uneven information access are therefore rarely properties of model outputs alone; instead, they emerge through configured interaction over the entire AI system pipeline [31, 34–36, 42]. Studying conversational AI therefore requires more than evaluating prompts or responses in isolation. It requires controlled human-subject studies that can manipulate specific parts of an interactive system while holding the rest of the experience constant.

In practice, however, constructing such studies is methodologically and technically demanding. Researchers must define and control a broader experimental setup shaped by the study procedure, task interface, agent behavior, and the surrounding context in which the interaction unfolds. Existing workflows typically require stitching together survey tools, bespoke systems, deployment infrastructure, and behavioral data collection. Commercial conversational systems further exacerbate this challenge because interface affordances, retrieval policies, and model behavior may change without notice. This leaves researcher with little control over the AI systems leading to creation of custom systems which hinder reproducibility and open-science. As a result, conversational AI studies are often costly to build, difficult to control, and hard to reproduce or build upon.

To address this gap, we present *Gricea*, a configurable research platform for authoring reproducible conversational AI studies. Gricea

provides controlled study authoring, task configuration, runtime instrumentation, and reusable open-science workflows within a single platform. It represents conversational-AI variants as explicit study artifacts that can be published, inspected, reused, and extended. It also lowers the technical overhead of designing and deploying controlled studies by allowing researchers to author study procedures and interactive task conditions without rebuilding surrounding infrastructure for each project.

Gricea views conversational AI studies as configurable, inspectable, and reusable research artifacts. Researchers can visually author multi-stage study procedures, configure interactive task conditions, deploy immutable study versions, and share published studies as reusable templates for replication and extension. More broadly, the platform supports an open-science workflow in which study configurations, prompts, public materials, and collected artifacts can remain visible to others, enabling auditing, reuse, and cumulative research in a rapidly evolving design space.

We evaluate Gricea through a study with $N = 10$ researchers and practitioners from diverse disciplinary backgrounds. Participants first completed an assisted authoring walkthrough using a fixed research question—*How does the stance of an AI assistant on controversial issues affect users’ perceived trust?*—and then designed their own conversational AI study in an unassisted think-aloud session. Through observed authoring behavior, task outcomes, post-task reflections, and interviews, we examine how Gricea supports researchers in authoring studies for diverse research questions on conversational AI.

This paper makes three contributions. **First**, we present Gricea, a platform that unifies study authoring, task-level control, participant-facing runtime execution, and fine-grained instrumentation. **Second**, we contribute platform mechanisms for reproducibility and open science, including immutable study versions, inspectable configurations, sharable links, and reusable community templates. **Third**, we report a formative evaluation showing how Gricea supports the authoring of conversational AI studies for diverse research questions by practitioners from diverse disciplinary backgrounds while revealing remaining needs for guidance, validation, and workflow support.

2 Formative Analysis: The Science of Conversational AI Studies

To scope the infrastructural requirements for Gricea, we conducted a formative design space analysis of papers on conversational AI systems. Our goal was to identify recurring patterns across prior work: what studies on conversational AI investigate, what they manipulate, how they are typically conducted, what technical demands those choices create, and what forms of infrastructure existing systems already provide. This section synthesizes those patterns to motivate the architecture of Gricea.

Analysis Procedure. We began by collecting 100 candidate papers using keyword combinations around *conversational AI*, *agent*, or *LLM*, together with terms related to *users*, *humans*, and *studies*, across venues and repositories such as CHI, UIST, ACM, and arXiv. We then filtered this set to 30 papers that centered participant-facing conversational or agentic AI systems and provided sufficient detail about the study design, system configuration, or evaluated

interaction condition. For each paper, we coded the study type, focus area, participant count, independent and dependent variables, between- and within-subject structure, procedural stages, system or pipeline components, analysis methods, and the overall structure of the study procedure. We then iteratively clustered these codes to identify recurring outcome areas, manipulation dimensions, procedural structures, and infrastructural demands.

As part of this analysis, we also examined how papers visually represented their study designs. Across the corpus, study procedures were often externalized through staged diagrams, branching structures, and flowchart-like depictions of condition assignment, tasks, and follow-up measures. This observation matters methodologically: researchers already represent study logic visually when communicating and planning their experiments. A platform for conversational AI studies should therefore support visual programming of studies rather than requiring researchers to reconstruct these structures manually in code.

2.1 What studies on conversational AI investigate

Studies on conversational AI investigate how configured assistants shape human behavior, judgment, and experience within particular task settings. Recent HCI work shows that this research space is already exceptionally broad. A review of 153 CHI papers found LLM-related research spanning communication and writing, augmenting human capabilities, education, responsible computing, programming, well-being and health, design, accessibility and aging, and creativity [29]. Studies on conversational AI occupy a substantial part of this space because conversational systems are increasingly deployed as interfaces for search, writing, tutoring, coding, decision support, and collaborative work.

Within these settings, the outcomes of interest are similarly broad. Prior work examines trust, reliance, persuasion, misinformation response, privacy behavior, writing quality, learning, and collaboration [20, 33–36, 42]. What links these studies is not a single application domain, but a common methodological concern: how a conversational system condition shapes what users believe, do, and produce over time. A research platform for this area therefore cannot be organized around one narrow task form such as chat alone.

2.2 The manipulation space of conversational AI studies

Our formative analysis shows that studies on conversational AI manipulate far more than prompts or underlying models. The true experimental object is a *configured interaction condition*. Prior conceptualizations often distinguish interface-level from agent-level manipulations; however, our analysis suggests that while these two categories remain central, they do not exhaust the condition being studied.

We synthesize the manipulation space into six interacting dimensions: *Interface Condition*, *Agent Condition*, *Context & Grounding*, *Task & Modality*, *Domain & Audience*, and *Study Procedure*. Table 1 summarizes these recurring dimensions. Interface conditions include layout, information presentation, and scaffolds that shape

what participants see and how they act [11, 12, 21]. Agent conditions dictate system instructions, stance, response style, tool-use policy, and initiative [13, 20, 26, 27]. Context and grounding determine what information the system can draw upon and how that evidence is surfaced [25, 33, 34]. Task modality, domain setting, audience, and study procedure further shape how participants interpret the interaction and how researchers measure its effects. Overall, studies on conversational AI must specify both what is manipulated and what is intentionally held constant across this entire space.

2.3 The Procedural Anatomy of Conversational AI Studies

The analysis also shows that studies on conversational AI are not only system manipulations; they are structured experimental procedures. A typical study may begin with consent, instructions, and pre-task elicitation, proceed through condition assignment and an interactive task, and conclude with post-task measures, reflection, or interview-based follow-up. Prior work repeatedly combines such elements, whether in survey-bracketed co-writing studies [14, 20], conversational search tasks with turn-level behavior and post-task attitudes [5, 33], or broader human–AI experiments that coordinate staged procedures with live interaction [30, 38]. A study on conversational AI is therefore simultaneously a *study procedure* and an *interactive runtime*.

The study procedure determines how participants move through the experiment: what they are told, what condition they receive, when branching occurs, and what measurements are collected before and after the task. The interactive runtime determines what the participant experiences within the task itself: what information the system receives, how it responds, how the interface adapts over time, and what actions and traces are logged. Because the final condition is produced jointly by both, a platform intended for studies on conversational AI must natively support the macro-procedure surrounding the task alongside the micro-runtime behavior within it.

A further observation from the corpus concerns how researchers already represent these procedures. Across the reviewed papers, study designs were frequently communicated through staged diagrams, branching depictions, and flowchart-like representations of condition assignment, tasks, and follow-up measures. This suggests that visual programming is not merely an interface preference. Researchers already externalize experimental logic in visual form when planning and communicating their studies. A platform for configurable conversational-AI studies should therefore support visual programming of procedures and task logic in a way that matches these existing practices.

2.4 Synthesis of Infrastructural Challenges

Executing studies across this broad manipulation space is challenging because the experimental condition is distributed across many components that must be built and controlled in tandem. Researchers often need to stitch together survey tools, custom interfaces, backend orchestration, model and retrieval pipelines,

Dimension	Typical manipulations in prior work	Why it matters for controlled studies
Interface Condition	Layout and workspace organization; citation and provenance display; highlighting; progress guides; verification prompts; structural overlays; editability of shared artifacts [11, 12, 21]	These choices determine what participants see, what actions feel available, and how evidence and control are distributed during the interaction.
Agent Condition	System instructions; persona or role framing; stance; response style; initiative; clarification behavior; trust-adaptive interventions; tool-use policy [13, 20, 26, 27]	These choices determine how the assistant behaves over time and what kind of partner, advisor, or information source the participant encounters.
Context & Grounding	Retrieved documents; retrieval policy; context filtering; citation constraints; external evidence injection; opposing or balanced context [25, 33, 34]	These choices determine what information the system can use and how the participant's informational environment is constructed.
Task & Modality	Search, writing, tutoring, coding, annotation, voice, multimodal interaction, and direct artifact manipulation [6, 18, 30, 38]	Task type and modality shape the meaning of the interaction, the actions available to users, and the outcomes that can be measured.
Domain & Audience	Topic domain, expertise level, prior beliefs, participant population, targeted community, and high-stakes versus low-stakes settings [19, 29, 31]	The same configured system can produce different behavior depending on who uses it, for what purpose, and in which domain context.
Study Procedure	Pre-task elicitation; condition assignment; branching; within- or between-subject structure; post-task measures; longitudinal follow-up [7, 20, 30]	The procedure determines how participants encounter the condition, what is held constant across sessions, and how outcomes are interpreted.

Table 1: Recurring dimensions in the manipulation space of studies on conversational AI, synthesized from the formative analysis. The configured interaction condition depends not only on agent behavior and interface design, but also on context, task setting, domain, audience, and study procedure.

assignment logic, deployment infrastructure, and fine-grained behavioral logging. Even when the intended manipulation is conceptually straightforward, implementing it as a controlled, reproducible participant experience requires steep engineering effort.

Recent infrastructure papers make similar observations in adjacent settings. Deliberate Lab argues that existing behavioral research tools remain poorly aligned with real-time, AI-mediated experiments and often require bespoke development for customization [30]. *Through the Lens* makes a related point for controlled studies of human-agent collaboration, arguing that the lack of an open, configurable platform makes it difficult to manipulate interaction controls across paradigms and to preserve them as reusable research artifacts [38]. The challenge for studies on conversational AI is closely related, but centered on conversational systems whose conditions depend jointly on procedural staging and interaction runtimes.

The opacity of commercial conversational systems makes this challenge sharper. Interface affordances, retrieval behavior, model versions, and system defaults may change without notice, placing some of the most critical experimental levers outside the researcher's control. The result is not that existing studies are unsuccessful—many are rigorously designed and highly informative. The problem is that each study carries a prohibitive implementation burden, and much of the resulting configuration remains embedded in one-off infrastructure rather than preserved as an explicit and reusable research artifact.

2.5 Existing Infrastructural Efforts and Remaining Gaps

Prior infrastructural efforts address important parts of this problem, but they do so from different starting points. One relevant lineage is general experiment infrastructure. LabintheWild, oTree, Empirica, and related systems focus on scalable online experimentation, participant routing, staged procedures, and deployment [2, 10, 30, 32]. These systems make it easier to run large-scale or structured studies, but they are not centered on participant-facing conversational conditions as configurable runtime artifacts. Their abstractions are strongest at the level of study procedure and deployment.

A second lineage focuses on lowering technical barriers for authoring or adapting computational behavior. TEA provides a high-level language and runtime that lets users express study design, assumptions, and hypotheses, then compiles them into valid statistical analyses [22]. Canvil shows that designers working with LLM-powered experiences need a translational mechanism that allows design requirements to shape and be shaped by model behavior [16]. ChainForge, PromptChainer, and related systems similarly reduce friction around prompt design, model comparison, and runtime experimentation [4, 8, 37]. These systems are highly relevant because they show how higher-level abstractions can lower technical barriers. However, they stop short of full study construction: they typically do not represent multi-stage procedures, participant assignment logic, publication/versioning, or reusable public study artifacts.

A third lineage consists of configurable environments for studying human-AI teaming and collaboration. *Through the Lens* is the

closest platform paper in genre and ambition because it presents a configurable, researcher-facing system for controlled human-agent collaboration studies [38]. CREW is also relevant as a platform for human-AI teaming research in real-time decision-making scenarios [41]. These systems show the value of configurable environments for studying interactive human-AI systems, but their center of gravity is collaboration and teaming rather than conversational-AI studies more broadly.

Taken together, these efforts reveal a consistent gap. Existing systems support procedural orchestration, higher-level authoring of computational behavior, or configurable human-AI environments, but not the full combination required for reproducible, configurable conversational-AI studies with lower technical barriers and reusable public artifacts. This gap motivates the design desiderata below.

3 Design Desiderata

The formative analysis above suggests that infrastructure for studies on conversational AI must satisfy requirements that go beyond either survey administration or interface prototyping alone. A useful platform must represent configured study conditions explicitly, support both the larger study procedure and the interactive task runtime, reduce the technical effort required to author such studies, and preserve the resulting artifact in a form that can be inspected and reused. Building on this analysis, we derive five design desiderata for Gricea.

D1: Explicit Representation of Study Conditions.

The platform must represent the configured interaction condition shown to participants rather than only isolated prompts, screens, or model calls. Studies on conversational AI are defined by the relation among procedure, interface, agent behavior, and contextual configuration. The infrastructure must therefore preserve these elements as part of the authored study artifact so that the condition itself remains legible.

D2: Coupled Support for Study Flow and Task Flow.

The platform must support both the macro-level structure of the study and the micro-level behavior of the interactive task. Study procedure determines how participants move through instructions, surveys, assignments, and post-task measures. Task runtime determines what the participant experiences within the interactive condition itself. Because studies on conversational AI depend on both, the infrastructure must support them jointly while preserving their distinction.

D3: Lower Technical Barriers to Controlled Study Authoring.

The platform must reduce the amount of bespoke engineering required to build and deploy controlled studies on conversational AI. Our formative study revealed that researchers often already externalize study logic through staged diagrams, branches, and condition flows during planning. The authoring model should align with these existing research practices so that researchers can move

from a study concept to an executable artifact without reconstructing surrounding infrastructure from scratch.

D4: Reproducibility Through Inspectable and Reusable Artifacts.

The platform must preserve authored studies as explicit research artifacts rather than leaving critical details embedded in transient setup steps or one-off implementations. Study versions, prompts, configuration logic, and public-facing materials must remain inspectable after deployment. This is necessary not only for reproducibility, but also for reuse. A published study should be understandable enough that another researcher can inspect it, run it, adapt it, and build on it.

D5: Extensibility Across Tasks, Modalities, and Study Settings.

The platform must remain extensible as the design space of conversational AI continues to expand. Studies may involve text chat, coding, multimodal interaction, voice, longer-running procedures, or community-facing workflows. The infrastructure must support extension across this broader space without requiring reimplementations of the surrounding system each time a new task type, modality, or study setting is introduced.

4 Gricea: A Platform for Configurable Conversational AI Studies

Gricea is a research platform for authoring, executing, and sharing controlled studies on conversational AI. The system architecture consists of three decoupled layers: a researcher-facing visual authoring environment, a participant-facing execution runtime, and a publication layer for versioning, reuse, and community sharing. Gricea represents each study as a structured artifact that specifies both the macro-experimental procedure and the micro-interactive task condition embedded within it. This representation allows complex study conditions to be authored, executed, preserved, and reused within a unified framework rather than reconstructed through ad hoc infrastructure.

The platform's architecture is organized around the conceptual distinction developed in the previous sections. Because a study on conversational AI combines a larger experimental procedure with an interactive runtime condition, Gricea operationalizes study designs as directed graphs. A *Study Flow* graph governs the macro-level routing of the experiment, while nested *Task Flow* graphs dictate the real-time behavior of the participant-facing interaction. Publication serializes these graphs, their configurations, and their assets into an immutable study version that the runtime engine executes directly.

4.1 Core System Abstractions

The platform represents authored studies through a strict taxonomy of explicit system objects. *Study Nodes* and *Study Flow* encode the procedural state machine of an experiment. *Task Nodes* and *Task Flow* encode the event-driven logic of an interactive condition. *Logic Nodes* encode branching, variable assignment, randomization, and condition control. At publication time, these authored objects are

compiled into a *Published Study* and a corresponding *Study Version*. To support open science, the system further maintains reusable data structures such as *Templates*, *Community Document Collections*, and *Participant Analytics*.

Study Flow. The Study Flow graph encodes the procedural structure through which participants navigate. Its nodes represent stages such as instructions, surveys, task entry, branching, randomization, within-subject blocks, annotations, and completion states. Operating as a macro-level state machine, the Study Flow engine resolves participant progression across these nodes by evaluating assignment logic, localized variable states, and procedural dependencies.

Task Flow. The Task Flow graph encodes the runtime behavior of the interactive condition. Its nodes represent operations such as LLM inference calls, retrieval and context injection (RAG), real-time voice handling, looping structures, and UI component updates. The Task Flow engine orchestrates these operations during execution, binding authored configurations to runtime events and component states. It serves as the execution model for the conversational condition, rather than acting merely as a container for a static chat interface.

Logic and Condition Control. Condition control is embedded directly into the graph topology. Logic Nodes, variable assignments, deterministic randomization algorithms, and within-subject routing blocks specify exactly which variables fluctuate across participants and which remain fixed. This architectural choice makes the manipulated and controlled variables explicit within the platform's representation, rather than leaving them implicit in external orchestration code.

4.2 Researcher Authoring Environment

Researchers author studies visually through node-based canvas editors for both Study Flow and Task Flow. By externalizing logic into a visual node-graph, the authoring environment closes the gap between conceptual experimental design and executable infrastructure. The model mirrors how controlled studies are mapped in practice—via staged procedures, condition branches, and task entry points—allowing researchers to construct the procedure and the interactive condition within the same environment while preserving their structural boundaries.

At the study level, researchers define the procedural scaffolding: instructions, elicitation stages, branching logic, randomization, within-subject ordering, and task entry points. At the task level, they configure the interactive parameters: system prompts, retrieved context, model hyperparameters, tool access policies, UI layout, scaffolds, and modality-specific settings.

Crucially, the platform separates iterative authoring from deployment. Researchers can preview and revise drafts in a mutable state. Publication, however, serializes the authored graphs, configuration states, prompts, and runtime bindings into a version-locked study artifact. The participant runtime strictly executes this published version. This immutability is central to Gricea's reproducibility guarantees, ensuring the exact condition deployed to participants is permanently preserved.

4.3 Participant Runtime and Instrumentation

The participant runtime acts as the client-side execution engine, rendering the assigned condition to each participant based on the published artifact. It resolves the Study Flow for procedural progression, invokes the associated Task Flow upon task entry, and binds participant interactions to the authored graph logic. The runtime serves as a dynamic renderer for the serialized study, rather than a hard-coded frontend application.

The runtime natively supports a broad taxonomy of conversational AI settings, including standard chat, web-based search tasks, voice interaction, collaborative coding, annotation workflows, image-generation, and split-view configurations. Because these tasks share a unified underlying execution model despite differing in modality, diverse conversational conditions can be deployed while remaining legible to the platform's data infrastructure.

Furthermore, the runtime embeds a high-resolution telemetry pipeline. Depending on task configuration, the platform autonomously logs timestamps, clickstreams, component focus events, scroll depths, keystroke edits, and task-specific interaction traces. Where enabled, the system captures voice recordings, screen recordings, file upload traces, and eye-tracking integration. Because this telemetry is strictly bound to the immutable Study Version, participant behavior can be definitively analyzed relative to the exact condition rendered.

4.4 Publication, Reuse, and Community Workflows

Gricea treats reproducibility as an intrinsic property of the study artifact itself. Published studies are archived as immutable versions with fully inspectable configurations. Prompts, task parameters, graph structures, assignment logic, and public materials remain permanently accessible, preventing experimental methods from degrading into unrecoverable legacy code. Published studies can be distributed via shareable links, and robust template cloning allows prior studies to serve as immediate starting points for subsequent research.

The publication layer extends this paradigm into a broader open-science ecosystem. Researchers can publish *Community Templates*, share *Community Document Collections* used for RAG grounding, and release de-identified dataset assets. In this capacity, the publication layer does not simply archive output data; it preserves the functional, authored condition as a reusable research object.

Consequently, a published study functions as more than a deployed interface. A peer researcher can inspect the underlying graph logic, experience the study live as a participant, and fork the artifact to run an independent replication. This mechanism operationalizes open science: studies remain explicit, inspectable, and executable indefinitely.

4.5 Architecture and Extensibility

The platform is engineered as a decoupled, modular system spanning the visual authoring client, the execution runtime, and the backend data/publication workflows. This modularity allows the system to ingest new task types, UI components, and modality settings without requiring the surrounding procedural infrastructure

to be rebuilt. The architectural objective is not merely feature accretion, but the preservation of a unified schema for study artifacts across an evolving design space.

This extensibility is critical. Future conversational AI research will rapidly expand into richer multimodal environments, longitudinal deployments, and broader community-facing interventions. Gricea is architected to accommodate these frontiers within its existing paradigm of Study Flow, Task Flow, immutable serialization, and unified telemetry.

The following section evaluates whether these abstractions successfully translate to researchers in practice. Specifically, it examines whether Gricea empowers researchers from varied disciplines to internalize the platform's mental model, author rigorous studies across diverse conversational AI questions, and generate reproducible experimental artifacts.

5 Evaluation

5.1 Evaluation Rationale and Questions

Gricea is a research platform intended to support the authoring of many different kinds of conversational AI studies. Evaluating such a platform therefore requires more than asking whether users can complete a single predefined task. Prior work on HCI toolkit research has shown that systems contributions often require mixed evaluation strategies, since their value lies partly in what they enable others to build rather than only in the direct usability of a fixed interface [24, 28]. We adopt the same general principle for our evaluation.

Our evaluation focuses on three questions. **RQ1**: can authors from diverse background author studies with Gricea?; **RQ2**: Can Gricea accommodate a diverse set of research studies; **RQ3**: what frictions remain in the current authoring workflow?

5.2 Participants

We recruited $N = 10$ participants from diverse disciplinary backgrounds and roles, including researchers, students, faculty, and industry practitioners. The participant summary is provided in the Table 2. Of the 10 participants 5 self-identified as Male and 5 self-identified as female. The median age of participants was in the range of 25-34. There were 4 PhDs, 2 Master students, 2 professionals, 1 professor and 1 undergraduate student in our sample. Participants varied in their prior experience with controlled studies and in the kinds of conversational AI questions they had previously explored or hoped to explore. This diversity was intentional since Gricea is meant to onboard researchers from diverse backgrounds by lowering the barriers to conduct user studies.

5.3 Study Design and Procedure

Each session was designed to assess both onboarding and open-ended study authoring. The procedure drew on a common pattern in toolkit and platform evaluation: a structured task that helps participants acquire the system's basic mental model, followed by a more open-ended task that reveals how they apply the platform to questions that matter to them [4, 39]. Sessions consisted of four phases: pre-task interview, assisted authoring walkthrough, open-ended think-aloud authoring, and post-task reflection. The total study lasted 90 minutes and participants were paid 30\$ through

amazon gift cards. Throughout the session we recorded the participant screen and audio after obtaining their informed consent.

We began each session with a brief introduction followed by a pre-task interview on their background, prior experience with human subject studies, challenges encountered in running human subject studies, and research questions they were interested in the area of Conversational AI.

In the assisted authoring walkthrough, participants implemented a fixed research question: *How does the stance of an AI assistant on controversial issues affect users' perceived trust?* During this phase, the facilitator provided guidance on the platform features, while participants retained control of the interface and performed the authoring themselves.

In the open-ended phase, participants were asked to design a conversational AI study of their own choosing using Gricea. They were instructed to think aloud as they worked.

After the authoring tasks, participants completed a post-task survey and took part in a semi-structured interview. These instruments focused on perceived usability, expressivity, reproducibility, likely time savings relative to current workflows, and adoption potential. The interview further probed where participants felt confident, where they felt stuck, and what forms of support would make the platform more useful in their own work.

5.4 Analysis Approach

We analyzed the survey, transcripts and screen recordings using a combination of descriptive summaries and qualitative analysis. Completion, timing, hints, and breakdowns were summarized descriptively across participants. Think-aloud transcripts, facilitator notes, and interview responses were analyzed thematically to identify recurring patterns related to learnability, expressivity, reproducibility, and unmet support needs.

5.5 Findings

5.5.1 Overcoming the Prototyping and Reproducibility Bottleneck. In the pre-task interviews, participants consistently described current conversational AI workflows as bespoke, costly, and highly unstable. P7 noted, "There's no standardized way of conducting these studies... these models keep changing over time." P6 described building custom interfaces from scratch, noting, "I manually coded all of that," which took "a few weeks." For participants with limited programming experience, this overhead was prohibitive. P9 stated, "I cannot code so I never thought I would be able to do this on my own, I would have to just pay someone to build it out for me." Furthermore, reproducibility emerged as a primary concern. P7 articulated this explicitly: "If I want to release my system, it is not clear to me how my systems can stand over time and how will my study be replicated and built upon."

Following the authoring tasks, participants overwhelmingly viewed Gricea as a structural solution to these bottlenecks. P8 estimated that utilizing Gricea would save "at least 3 to 6 months of effort" and roughly "\$25,000 worth of money," noting that building their protocol manually "would have been a nightmare." This qualitative enthusiasm was supported by the post-task survey, where participants indicated a strong likelihood to recommend Gricea to colleagues ($M = 6.8$, $SE = 0.13$) and expressed high confidence that

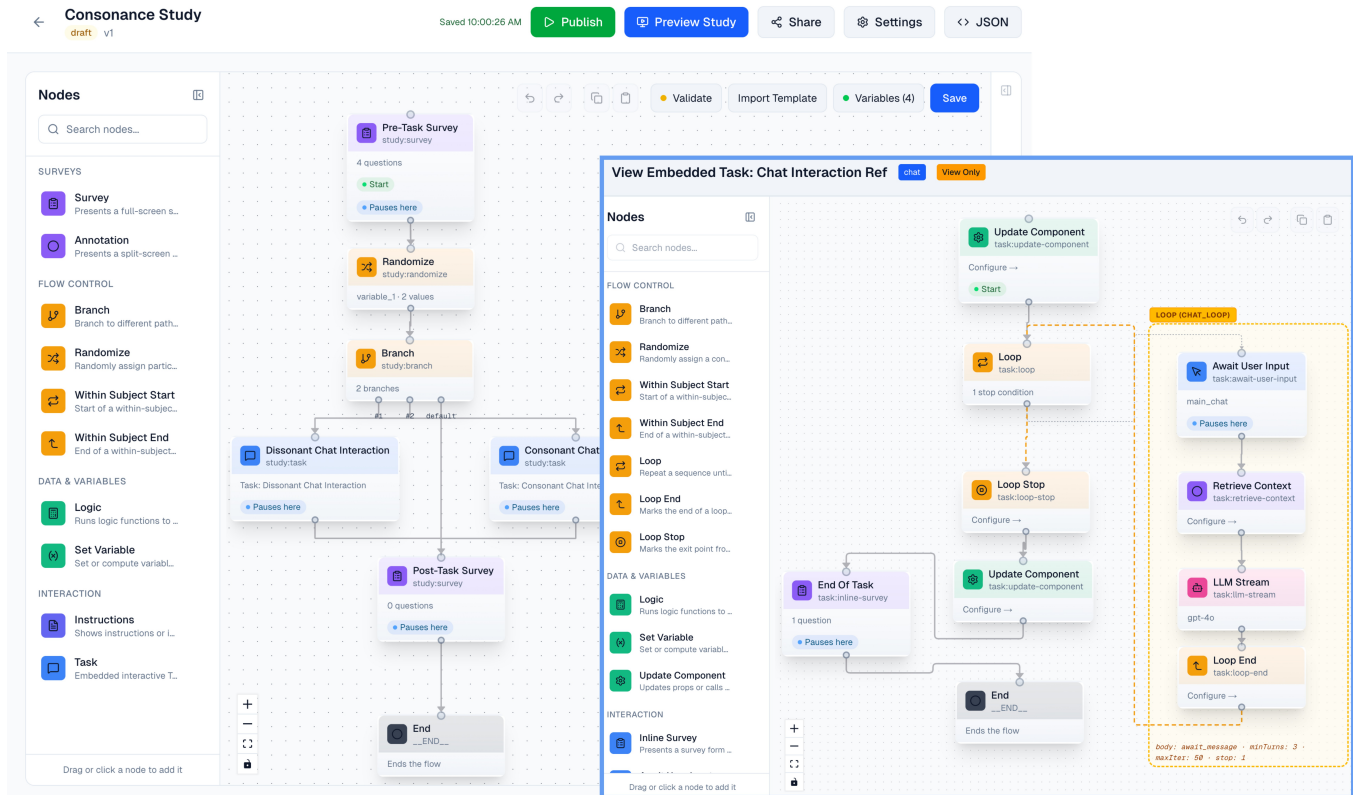


Figure 2: The Gricea authoring interface, illustrating how researchers can assemble multi-stage study procedures (left) and independently configure agent pipelines and interface scaffolds for specific conversational tasks (right).

studies authored in Gricea could be reliably reproduced by other researchers ($M = 5.9, SE = 0.37$).

5.5.2 Participants from diverse backgrounds were able to author studies. Participants across disciplines successfully mapped their conceptual study designs onto Gricea’s dual-flow abstractions, reporting high overall ease of use ($M = 6.1, SE = 0.31$). Several explicitly stated that the node-based architecture mirrored their internal cognitive models of experimental design. P1 noted that “all those component building blocks make sense to me” and could be used “intuitively” to build procedures. P3 found that “the plug, play, click and edit pipeline is really intuitive,” preserving their “chain of thought,” while P4 highlighted that “the most intuitive part is when you connect those together.”

Three platform strengths consistently emerged as critical for supporting researchers from such diverse backgrounds. First, the visual canvas provided necessary architectural clarity; P8 noted, “The visual interface is quite easy to develop the protocol... The visual overlay made a lot of difference.” Second, participants valued the unified consolidation of procedure and runtime. P6 appreciated having surveys, AI tasks, and deployment “contained into one system,” contrasting it with fragmented legacy workflows. Third, the localized validation and preview mechanisms were highly praised. P7 appreciated “being able to preview it... and seeing if what I thought is what is actually happening,” while P3 praised “the ease

with which a study could be validated,” calling the feature “super cool.” These features matter beyond basic usability; they render experimental intent inspectable and debuggable prior to deployment.

5.5.3 Gricea supported RQs across the Conversational AI Design Space. The open-ended phase provided robust evidence of Gricea’s expressive capacity. Participants successfully generated and operationalized a highly diverse set of research questions, confirming they were able to express the exact study designs they had in mind ($M = 5.5, SE = 0.40$).

Implemented studies spanned the breadth of the conversational AI manipulation space. P7 investigated how AI assistance affects subsequent non-AI creativity. P5 explored the “AI penalty” by testing evaluations of text with and without AI-disclosure. P2 operationalized a study on serendipity across standard web search, generative search, and RAG architectures. Other participants configured studies on proactive versus reactive coding agents (P3), LLM-augmented debates with human moderators (P8), conditions prompting users to utilize agentic tool-calls (P1), human fact-checking behaviors with generative AI (P9), and how different tiers of models exacerbate the digital divide (P10). The exact RQs are in Appendix 3.

Participants explicitly confirmed that Gricea easily accommodated these designs. P3 stated, “All of it was able to be done, and I have been trying to think of studies that I cannot run but I cannot think of one yet.” P6 reported Gricea supported “even more

than I thought was possible,” and P8 summarized the platform’s capacity as “all of it and more.” Even for highly domain-specific protocols—such as P6’s mechanical design collaboration or P8’s clinical protocols—participants found the platform broadly expressive enough to capture their required experimental manipulations and compliant with medical standards for PII data.

5.5.4 High Degrees of Freedom and the Need for additional scaffolding. While Gricea’s flexibility enabled a diverse RQs from diverse researchers, the high degrees of freedom also introduced new methodological frictions. Breakdowns occurred most often for variables and branching: P4 found the randomization logic “confusing for me for like the branching and the randomized [nodes],” while P1 struggled with “how to dynamically insert a prompt” using variables.

Relatedly, several participants experienced a conceptual boundary when transitioning from the macro *Study Flow* to the micro *Task Flow*. P6 noted that entering the Task Builder “brought us to another workflow that was a little confusing.” P2 found the task canvas overly granular, stating that “it offers too much detail” and “for social scientists who don’t need complex configurations they might to abstract away the complexities.” These frictions suggest that while Gricea’s core abstractions are powerful, researchers especially those from non CS background require stronger systemic scaffolding to manage the flow of variables across nested nodes. P1 and P4 also noted a “high-learning curve” to use the platform and all of it’s features.

To resolve these issues, participants requested additional support for *methodological mapping*. P7 desired an AI feature to provide “a basic starting template” to help structure the flow. P1 articulated an important boundary condition for the platform: Gricea is highly effective when the experimental design is already concrete, but early-stage ideation still requires “collaborative brainstorming with AI agents.” All participants however echoed that once they were used to the platform and ran a few studies they could see themselves getting over these barriers.

6 Discussion

6.1 Conversational AI Studies as Research Artifacts

A central implication of this work is that studies on conversational AI should be preserved as explicit research artifacts. In many current workflows, the condition shown to participants is fragmented across study procedures, custom interfaces, prompts, contextual grounding, and external API orchestration. Only part of this condition is typically visible in the final publication, while much of the actual manipulation remains buried in ad hoc code, undocumented defaults, or transient platform configuration. Gricea addresses this problem by preserving the authored study as a visible artifact rather than treating the deployed interface as the only durable outcome of the research process.

The evaluation suggests that this visibility matters to researchers. Participants repeatedly emphasized that versioning, inspectability, reusable templates, and sharable links are valuable not only as workflow conveniences, but as mechanisms for understanding exactly what another researcher built and ran. In this sense, the value of the

platform is partly epistemic. By preserving the authored condition in a form that others can inspect, rerun, and fork, Gricea makes methods easier to audit, interpret, and build upon.

This implication connects directly to broader concerns about reproducibility in LLM-based HCI and behavioral research. Recent reporting frameworks such as GUIDE-LLM argue that studies involving large language models require more explicit documentation of prompts, model settings, privacy handling, and validation procedures [17]. Work in HCI has similarly emphasized that reproducibility depends not only on reporting outcomes, but also on supporting the practices through which methods are made inspectable and reusable [15]. Gricea complements these efforts by shifting part of that burden from post-hoc reporting to the structure of the study artifact itself. This position also aligns with the broader argument of *Experiments as Code*, which frames reproducibility as a property of how experiments are represented and managed rather than as an afterthought of documentation alone [1].

6.2 Study Flow and Task Flow as a Methodological Abstraction

A second implication concerns the distinction between *Study Flow* and *Task Flow*. In Gricea, this distinction appears as a platform abstraction. More importantly, it also clarifies the methodological structure of studies on conversational AI. Such studies are not only procedures that assign participants to conditions, collect pre- and post-task measures, and manage staged progression. They are also interactive runtimes in which participants encounter a configured conversational condition that unfolds turn by turn. The participant experience depends on both.

The formative analysis already pointed toward this distinction. Across the reviewed papers, study procedures were often communicated through staged diagrams, branching depictions, and flowchart-like structures. These diagrams did not merely summarize a paper after the fact; they externalized how researchers already conceptualize and communicate experimental logic. The evaluation reinforced this point. Participants repeatedly described Gricea’s visual canvas as matching how they thought about study design, and several highlighted that connecting stages and previewing the resulting flow helped them reason about the experiment more concretely. Visual authoring therefore matters not only because it is usable, but because it aligns with how study logic is already represented in practice.

This abstraction also helps explain why studies on conversational AI fit awkwardly within existing infrastructure. General experiment platforms such as oTree and Empirica are strong at procedural staging, participant management, and assignment logic [2, 10]. LLM-facing tools such as ChainForge are strong at prompt, model, and runtime experimentation [4]. Human-agent research platforms such as *Through the Lens* move closer to configurable interaction studies [38]. What studies on conversational AI require, however, is a way to represent both the larger procedure and the participant-facing runtime condition together. The distinction between *Study Flow* and *Task Flow* gives the paper a vocabulary for that requirement, and may remain useful beyond Gricea itself as a way of reasoning about conversational-AI research infrastructure.

6.3 Lowering Technical Barriers Requires Methodological Scaffolding

Lowering technical barriers in this area is not only a matter of convenience. It is also a research-methods contribution. When the cost of building a controlled conversational-AI study is high, the set of people who can run such studies narrows, and so does the set of research questions that are likely to be explored. Infrastructure overhead therefore shapes the research space itself.

Our evaluation provides early evidence for this point. Participants from different disciplinary backgrounds were able to formulate and begin implementing studies on trust, privacy disclosure, serendipity, tutoring modalities, political bias, coding assistants, fact-checking, creativity, and digital inequality. The significance of this result is not only that participants could use the interface. It is that the platform was expressive enough to support a range of research questions that would otherwise require substantial bespoke engineering before they could be translated into controlled studies.

At the same time, lowering technical barriers does not reduce the need for methodological rigor. It shifts more responsibility onto the platform to help researchers construct valid studies. This is where the strongest boundary condition from the evaluation appears. Participants did not primarily ask for more low-level control. Instead, they asked for better support in translating research intent into study logic: help with variables, branching, randomization, validation, and previewing how authored choices would shape the participant condition. Similar tensions appear in other researcher-facing systems, where making a tool easier to use also increases the need for guidance that supports appropriate methodological choices [4, 38]. In the context of Gricea, the main remaining challenge is therefore not representational power, but methodological scaffolding.

6.4 Open-Science Workflows for Cumulative Conversational-AI Research

The open-science contribution of Gricea should not be understood only as publication after the fact. In studies on conversational AI, openness requires preserving the configured runtime condition itself. Prompts, interface structure, assignment logic, contextual grounding, and public materials all shape the participant experience. If these elements remain hidden or ephemeral, then much of the study's method disappears even when the paper is public.

This is why templates, inspectable published studies, and sharable study links are central rather than secondary features. A reusable template is not simply a saved interface. It is a preserved study condition that another researcher can inspect, run, and adapt. The ability to run a published study as a participant before forking it is especially important in this regard, because it allows reuse to begin from the lived participant condition rather than from a textual description alone.

More broadly, the platform's community-facing layer suggests one path toward cumulative conversational-AI research. Replication, extension, and comparison become easier when prior studies remain available as explicit artifacts rather than ending as isolated prototypes plus a paper. Whether this possibility is fully realized will depend on longer-term adoption, but the platform is built

around the premise that studies on conversational AI should remain usable after the original deployment ends.

Taken together, these implications suggest that future infrastructure for conversational-AI research should not only support interaction design, but also explicit condition representation, runtime instrumentation, versioning, and reuse. The next section considers the present limits of this contribution.

7 Limitations

This work presents a formative evaluation of a research platform rather than a summative validation across all intended users and study settings. The user study provides evidence about how researchers from varied backgrounds used Gricea to author conversational AI studies, but the sample remains modest.

The evaluation also covers only part of the platform's implemented scope. Gricea supports a broader range of task types, modalities, publication workflows, and community-facing features than any single study could exercise. The present paper therefore validates the platform's core authoring model more directly than every implemented capability.

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.1 Participant Details

The participant summary is provided in table 2 and the detailed RQs that each participant came up with for the open-ended task are in Table 3.

ID	Role	Conducted studies?	Study count	Primary research interests / intended study
P01	Researcher	Yes	5–10	Capability boundary of conversational AI; what tasks can and cannot be automated; human–human / human–agent boundaries.
P02	Master Student	No	N/A	Serendipity in search; comparing web search, GenAI search, and RAG search; detailed search-session logging.
P03	Master Student	No	N.A.	Legal AI and hallucinated precedent; coding agents; mental-health advice and human influence.
P04	PhD	Yes	1–5	Social consequences of conversational agents; tutor-like interaction; transfer from human–AI to human–human interaction.
P05	PhD	Yes	1–5	AI penalty: how judgments change when people learn a text was AI-generated versus human-generated.
P06	PhD	Yes	1–5	Utility of AI systems in design / mechanical collaboration; AI personas in design tasks.
P07	PhD	Yes	1–5	Long-term creativity after AI assistance; comparing no-AI, AI-guidance, and answer-like AI support.
P08	Professor	Yes	10+	Surgeon knowledge assessment; using LLMs to identify knowledge deficits and support dialogue around experiential expertise.
P09	Professional	Yes	10+	How users use physical devices related to healthcare;
P10	Undergraduate Student	No	N/A	N/A

Table 2: Completed participant records used in the analysis ($n = 10$ usable completed records in the attached JSON export). Participants spanned multiple roles and brought diverse conversational-AI study ideas to the authoring task.

Participant Open-ended research question	
P01	Under what conditions and tasks do users rely on different agentic tool calls, such as web search?
P02	How does serendipity differ across web search, generative-AI search, and retrieval-augmented generation?
P03	How do proactive versus reactive coding agents shape participants' coding practices?
P04	How does AI-assisted creative writing compare to human-assisted creative writing?
P05	If people cannot distinguish AI-written text from human-written text, where does the perceived "AI penalty" in evaluation come from?
P06	How do different personas of AI chatbots affect engineers in mechanical design tasks?
P07	How does AI assistance in creativity tasks affect subsequent creativity without AI?
P08	How does an LLM-augmented debate with a human moderator affect discussion and decision-making?
P09	How do people fact-check information when using generative AI?
P10	How do differences in model tier exacerbate the digital divide?

Table 3: Open-ended research questions participants brought to the Gricea authoring task. The breadth of these questions illustrates the range of conversational-AI studies participants were able to formulate and begin implementing with the platform.